

Compact metric space.

$f: [a, b] \rightarrow \mathbb{R}$ is continuous then f attains its extreme value in $[a, b]$. This theory also works for closed & bdd interval.

$$f(x) = \frac{1}{x}, x \in (0, 1)$$

$$g(x) = x, x \in [0, \infty)$$

The topological explanation for this is that closed & bdd interval has a special property called compactness which makes the extreme value theory (and many other results) works.

open cover :-

Let (X, d) be a complete metric space. An open cover of X is a (possibly infinite) collection $\{V_i\}_{i \in I}$ of open subsets of X such that $X \subseteq \bigcup_{i \in I} V_i$. We say that X is compact if every open cover has a finite subcover. i.e., $\exists$ finite $J \subset I$ s.t. $X \subseteq \bigcup_{j \in J} V_j$.

Ex:- ① $\mathbb{R}$ is not compact

$\Rightarrow$ Consider $\mathbb{R}$ with the usual metric. The space has many open cover i.e., we can write $\mathbb{R} \subseteq \bigcup_{x \in \mathbb{R}} B(x, 1)$. i.e. $\{B(x, 1) | x \in \mathbb{R}\}$ is an open cover of $\mathbb{R}$. Some of these balls are redundant because $B(1.9, 1)$ & $B(2.1, 1)$ are in the open cover so the ball $B(2, 1)$ is unnecessary and could be removed.

But no matter how many redundant balls you removed, you still need an infinite no. of balls in the collection to cover $\mathbb{R}$. because one could never cover $\mathbb{R}$ with finite no. of balls. Thus, $\mathbb{R}$ is not ~~complete~~ **COMPACT**.

② Show that $\{1, 2, 3\}$ is compact.

③ $(0, 1]$ is not compact in $\mathbb{R}$.

④ Any finite set in a metric space is compact.

Let E be compact subset in a metric space is compact (X, d) . ^{Then} E is complete, closed & bdd.

× Th. A set in $\mathbb{R}^n$ with the usual metric is compact iff it is closed & bdd.

Every closed subset of compact metric space is compact.

× # A set E in a metric space (X, d) is compact iff every infinite subset of E has at least one limit point in E .

$(0, 1]$ is not compact in $\mathbb{R}$

$\Rightarrow U_n = (\frac{1}{n}, 2)$ cover $(0, 1]$

We assume that U_n has a finite subcover F

F contains an element U_k ; $k \geq i$, $U_i \in F$.

$U_k \supseteq U_i$ $U_i \in F$.

$U F = U_k = (\frac{1}{k}, 2)$ ~~$\neq (0, 1]$~~ Thus, $(0, 1]$ is not contained in $U F$.

$\therefore (0, 1]$ is not compact, by definition.

× Th. A set in $\mathbb{R}^n$ with the usual metric is compact iff it is closed & bdd.

$\Rightarrow$ Let E be a subset in $\mathbb{R}^n$.

We know that if E is compact then it is closed & bdd.

Now we will show that converse.

Since E is bounded in $\mathbb{R}^n$. We can write $E \subseteq [-N, N] \times \dots \times [-N, N]$ (n times) for some $0 < N \in \mathbb{R}$, N is finite.

Since $[-N, N]$ is compact in $\mathbb{R}$, therefore $\prod_{r=1}^n [-N, N]$ is also compact in $\mathbb{R}^n$.

Since E is a closed subset of the compact set $[-N, N]^n$. Hence E is compact.

Sequentially compact :-

A metricspace (X, d) is sequentially compact if every sequence of points in X has a convergence subsequence converging to a point in X .

Ex:- $(0, 1]$ is not sequentially compact whereas $[0, 1]$ is sequentially compact.

Defⁿ:- Bounded and totally bounded metricspace.

2016 A metricspace (X, d) is said to be bounded if there is some open ball $B(x, r) = \{x \in X : |x - y| < r \ \forall y \in X\}$ which contains X entirely.

A metricspace (X, d) is said to be totally bounded if for every $\epsilon > 0$ X can be covered by a finite number of open balls of radius ϵ .

For instance $\mathbb{R}$ is not bounded. (It cannot be enclosed inside an open ball) and it is not totally bdd also.

But the set $[0, 1]$ is both bounded & totally bdd. as for any $\epsilon > 0$ $[0, 1]$ can be covered by a finite no.s of ϵ balls. Consider the set of integer $\mathbb{Z}$ with discrete metric $d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases}$

In other words all the integers are at distance 1 apart from each other under this metric.

In this case $\mathbb{Z}$ is bdd [as it is contained in $(0, 2)$]
 But it is not totally bdd because it is not possible to cover $\mathbb{Z}$ by finite no.s of open balls of radius $1/2$.

Ex:- Show that every totally bdd metricspace is bdd but converse is not true.

$\Rightarrow$ Let (X, d) be a totally bdd metricspace.

By definition. It can be covered by a finite nos. of open ball of radius 1. (say)

Let us called this ball as $B(x_1, 1), B(x_2, 1), \dots, B(x_n, 1)$

Let R denote the radius. $R = 1 + \max_{1 \leq i \leq n} d(x_i, x_1)$

from triangle inequality we see that

$$B(x_i, 1) \subset B(x_1, R) \quad \forall i$$

since $B(x_i, 1)$'s cover X . Thus we see that $X \subset B(x_1, R)$

$\Rightarrow X$ is bdd.

converse is not true.

consider the set of integers $\mathbb{Z}$ with discrete metricspace

$$d(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

In this case $\mathbb{Z}$ is bdd as it is contained in $(0, 2)$. But it is not totally bdd because it is not possible to cover $\mathbb{Z}$ by finite nos of open balls of radius $1/2$.

Ex:- Let (X, d) be a discrete metricspace with X infinite.

Then any open cover of X has no finite subcover.

If any set $\{x\} : x \in X$ is deleted then the result is a sub-cover of $\{x\} : x \in X$ no longer remains cover for X as X is infinite.

Th A discrete metricspace (X, d) is compact iff X is finite.

Ex:- In a metricspace (X, d) (i) Union of no of compact set is compact. (ii) Intersection of any collection of compact sets is compact.

2) ~~2014~~ Show that $(\mathbb{R}, d)$, $d(x, y) = \min\{1, |x - y|\}$ is ^{not} totally bdd
 $\Rightarrow$ Since $(\mathbb{R}, d)$ is bdd as $d(x, y) \leq 1 \forall x, y \in \mathbb{R}$.
 Let $A \subseteq \mathbb{R}$ be any finite set and also let x be the largest element in A .

Then $d(x+1, y) = 1$.

and so there is no balls of radius $1/2$ for $\mathbb{R}$.
 Thus $(\mathbb{R}, d)$ cannot be cover by finite no. of open balls of radius $1/2$.

Th A discrete metricspace (X, d) is compact iff it is finite.

$\Rightarrow$ Assume that X is compact, then every open cover of X has a finite subcover. Consider the open cover $\mathcal{U} = \{B(x, \epsilon) : x \in X\}$ for $0 < \epsilon < 1$. Then $\exists x_1, x_2, \dots, x_n$ such that $X \subset B(x_1, \epsilon) \cup B(x_2, \epsilon) \cup \dots \cup B(x_n, \epsilon)$. But $B(x_i, \epsilon) = \{x_i\}$ for $i = 1, \dots, n$ because we have discrete metric. So $X \subseteq \{x_1, x_2, \dots, x_n\}$ and hence X is finite.

Conversely assume X is finite, i.e. $X = \{x_1, x_2, \dots, x_n\}$.

Now let $\mathcal{U} = \{U_\lambda : \lambda \in \Lambda\}$ be any open cover for X .

This means that $x_i \in U_{\lambda_i}$ for some $\lambda_i \in \Lambda \forall i = 1, \dots, n$.

So, $X \subseteq U_{\lambda_1} \cup U_{\lambda_2} \cup \dots \cup U_{\lambda_n}$ and so we have produce a finite subcover and hence X is compact.

$\times$ Th^o - Unions of finite no.s of compact set is compact.

$\Rightarrow$ Let G be an open cover of F . Then G is an open

let $F_1, F_2, \dots, F_n$ be compact set.

$$F = F_1 \cup F_2 \cup \dots \cup F_n$$

Since F_i $i = 1, 2, \dots, n$ is compact, ~~it has~~ F_i has finite open subcover G_i . $i = 1, 2, \dots, n$.

$$\text{let } G_0 = G_1 \cup G_2 \cup \dots \cup G_n$$

G_0 is then a finite open subcover of F .

$\therefore F$ is compact.

× Th^o:- Prove that any collection of compact set is compact.

=> let $B_1, B_2, \dots$ be compact set.

For some countable set I enumerating a collection of closed sets B_i , we have that $\bigcap_{i \in I} B_i$ is closed, since the intersection of any collection of closed sets is also closed.

Then consider a sequence of finite subsets of I , J_n and elements of I , k_n , defined as follows:

$$J_1 \subset I; k_1 \in I, k_1 \notin J_1$$

$$J_2 = J_1 \cup \{k_1\}, k_2 \in I, k_2 \notin J_2$$

!

$$J_n = J_{n-1} \cup \{k_{n-1}\}, k_n \in I, k_n \notin J_n$$

Then for each $n \in \mathbb{N}$, $\bigcap_{i \in J_{n-1}} B_i$ is closed, and $B_{k_{n-1}}$ is a compact set (by construction). Since the intersection of a closed set & a compact set is also compact,

$(\bigcap_{i \in J_{n-1}} B_i) \cap B_{k_{n-1}} = \bigcap_{i \in J_n} B_i$ is also a compact set.

Since this holds true for all $n \in \mathbb{N}$, $\bigcap_{i \in I} B_i$ is compact. For any arbitrary collection of compact set $\{B_i\}_{i \in I}$.

Defⁿ:- (Finite intersection property)

A collection of closed set $\{C_i: i \in I\}$ (I is an index set), has the finite intersection property if every finite subcollection of the above collection has non-empty intersection.

The Borel-Lebesgue theorem:-

For an metric space (X, d) the following are equivalent- (i) X is compact

(ii) Every collection of closed subset of X having the finite intersection property has non-empty intersection.

(III) X is sequentially compact.

(IV) X is complete and totally bdd.

Th.

For a metricspace (X, d) X is compact iff "Every collection of closed subset of X having the finite intersection property has nonempty intersection."

Let X be compact and consider a collection of closed subsets $\{C_i : i \in I\}$ of X having finite intersection property. We need to show that $\bigcap_{i \in I} C_i \neq \emptyset$

By contradiction we assume $\bigcap_{i \in I} C_i = \emptyset$.

put $U_i = C_i^c \forall i \in I$

Then each U_i is open

Then $\{U_i : i \in I\}$ is an open cover of X , since $\bigcup_{i \in I} U_i = \bigcup_{i \in I} C_i^c$

~~Since~~

$$= \left(\bigcap_{i \in I} C_i \right)^c = \emptyset^c = \emptyset$$

- As X is compact ~~every~~ $\exists$ a finite subcover say $\{U_{n_1}, U_{n_2}, \dots, U_{n_k}\}$ of X . Therefore $X = U_{n_1} \cup U_{n_2} \cup \dots \cup U_{n_k}$

$$= (U_{n_1}^c \cap U_{n_2}^c \cap \dots \cap U_{n_k}^c)^c$$

$$= (C_{n_1} \cap C_{n_2} \cap \dots \cap C_{n_k})^c$$

$$\Rightarrow \emptyset \in C_{n_1} \cap C_{n_2} \cap \dots \cap C_{n_k} = \emptyset$$

which is clearly a contradiction of the collection $\{C_i : i \in I\}$ that has FIP. Hence $\bigcap_{i \in I} C_i \neq \emptyset$

Conversely, suppose that $\{U_i : i \in I\}$ is an open cover of X put $C_i = U_i^c \forall i$

Suppose further that no finite subcollection of $\{U_i : i \in I\}$ covers X . Then if a subcollection $\{C_{n_1}, C_{n_2}, \dots, C_{n_k}\}$ of $\{C_i : i \in I\}$

satisfies $C_1 \cap C_2 \cap \dots \cap C_k = \emptyset$, we could have

$$U_1 \cup U_2 \cup \dots \cup U_k$$

$$= (U_1^c \cap U_2^c \cap \dots \cap U_k^c)^c$$

$$= (C_1 \cap C_2 \cap \dots \cap C_k)^c = \emptyset^c = X, \text{ which is a contradiction}$$

on with the assumption that no finite subcollection of U_i covers X . Thus $\{C_i : i \in I\}$ has ~~an~~ FIP.

Hence we can write $\bigcap_{i \in I} C_i \neq \emptyset$.

$$\Rightarrow \bigcap_{i \in I} U_i^c \neq \emptyset$$

($\forall i \in I$)

$$\Rightarrow \bigcup_{i \in I} U_i \neq X$$

which violates the assumption that the collection $\{U_i : i \in I\}$ is an open cover of X .

Hence X is compact.